

CS689 Final Project Delta Report

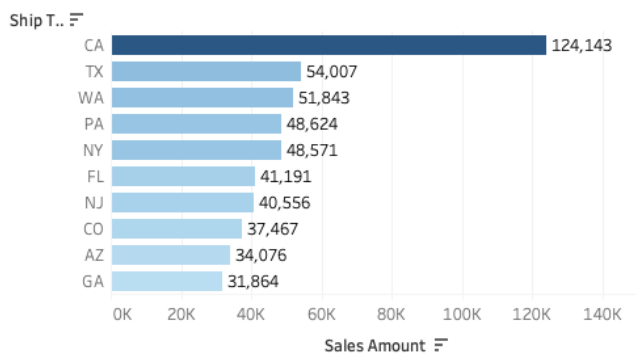
Student: Susan Jia

Course: MET CS689 Designing and Implementing a Data Warehouse

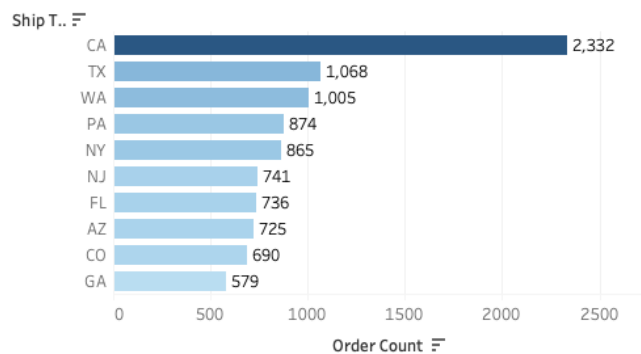
Project Title: Amazon Order & Revenue Data Warehouse

Amazon Order & Revenue Performance Dashboard

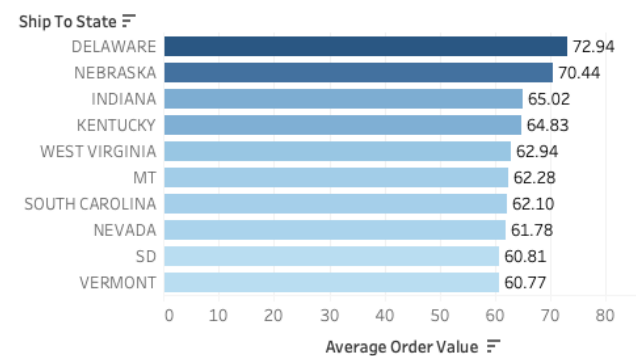
Top Destination States by Sales



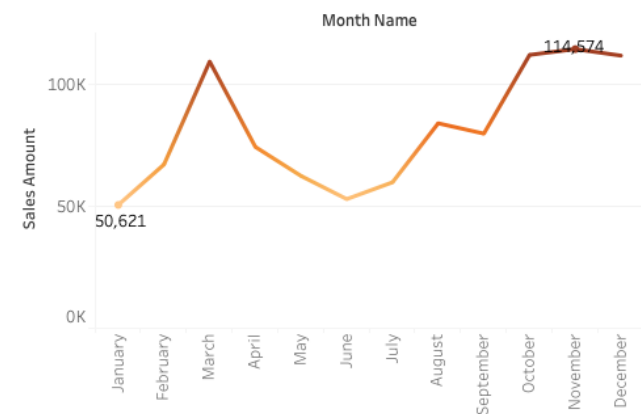
Order Volume by State



Average Order Value by State



Monthly Revenue Trend



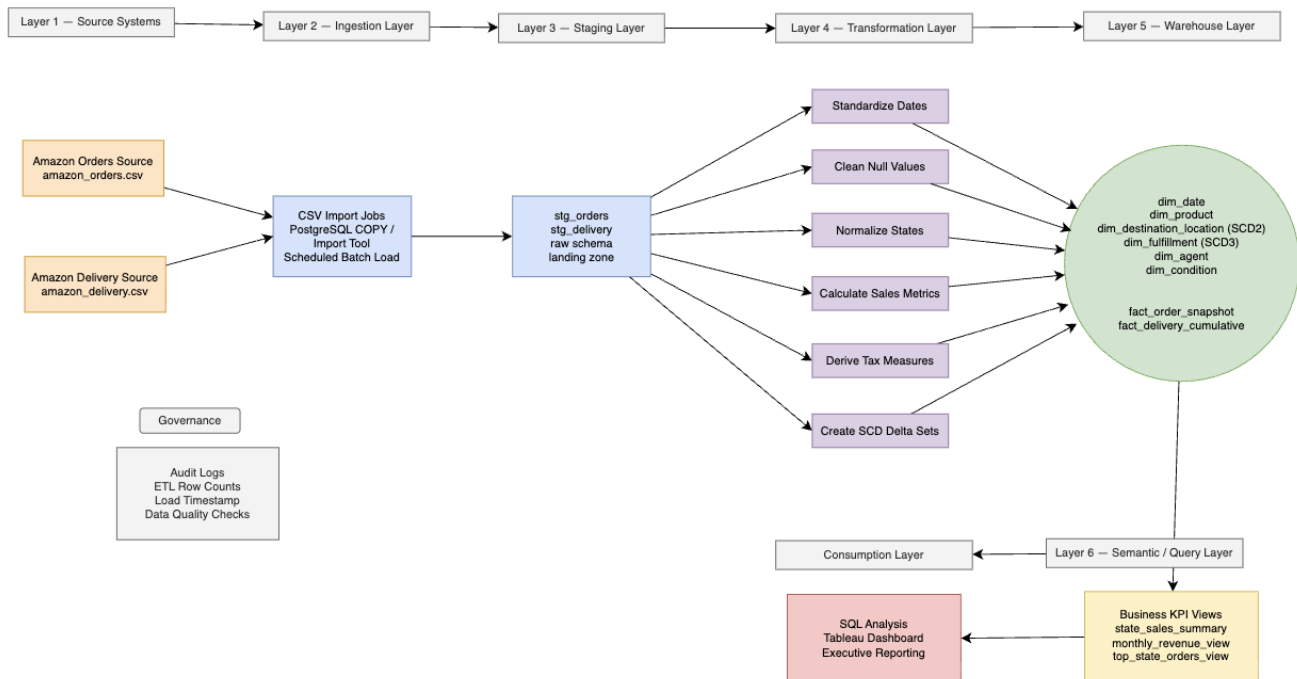
1. Overview

This Delta Report summarizes the final improvements and updates completed after the project presentation.

The final version strengthens the warehouse architecture, ETL process, SQL analytics, Slowly Changing Dimension implementation, and Tableau dashboard design.

Additional enhancements were made to improve dimensional modeling clarity, business insights, chart variety, and overall presentation quality.

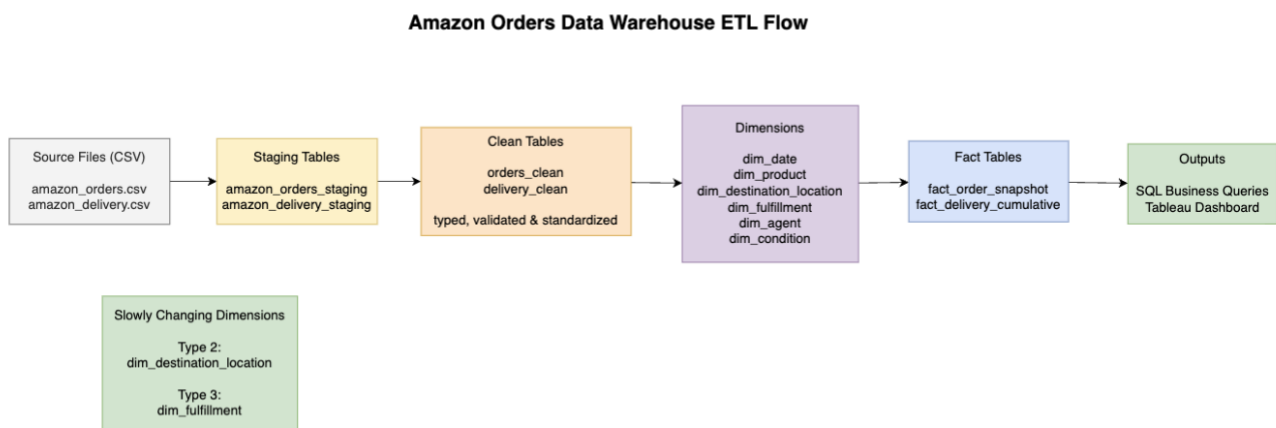
Amazon Commerce Data Platform Architecture



2. Improvements Made After Presentation

Based on the feedback received during the presentation, the following improvements were completed:

- Updated the dimensional model with clear labels
- Added fact table type labels
- Added SCD Type 2 and Type 3 labels
- Improved ETL flow diagram
- Added Clean Tables layer
- Improved dashboard titles and layout
- Added hierarchy drill-down
- Added dashboard filters
- Improved chart variety
- Refined business conclusions



3. Final Warehouse Design

Fact Tables

fact_order_snapshot

Type: Transactional Snapshot Fact

Grain: One row per order transaction

fact_delivery_cumulative

Type: Cumulative Fact

Grain: One row per delivery process record

Dimension Tables

dim_date

dim_product

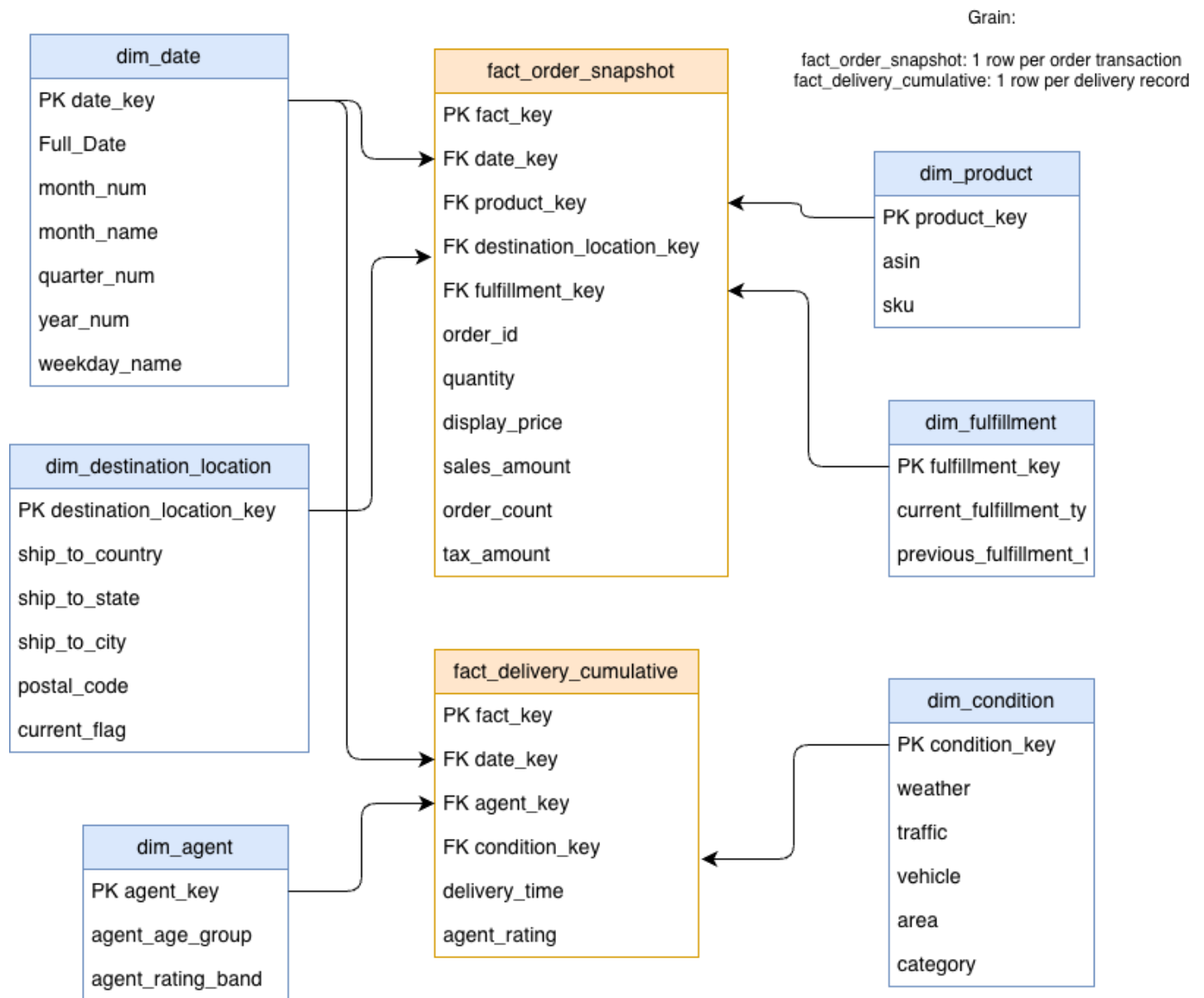
dim_destination_location (SCD Type 2)

dim_fulfillment (SCD Type 3)

dim_agent

dim_condition

Amazon Sales Warehouse Dimensional Model



SCD Type 2: dim_destination_location
SCD Type 3: dim_fulfillment

4. Business Question 1

Which states generate the highest total sales revenue?

```
4 select
5     dd.year_num,
6     dd.month_num,
7     dl.ship_to_state,
8     count(*) as order_count,
9     dense_rank() over (
10         partition by dd.year_num, dd.month_num
11         order by count(*) desc
12     ) as state_rank
13 from fact_order_snapshot f
14 join dim_date dd
15     on f.date_key = dd.date_key
16 join dim_destination_location dl
17     on f.destination_location_key = dl.destination_location_key
18 group by
19     dd.year_num,
20     dd.month_num,
21     dl.ship_to_state
22 order by
23     dd.year_num,
24     dd.month_num,
25     state_rank,
26     dl.ship_to_state;
```

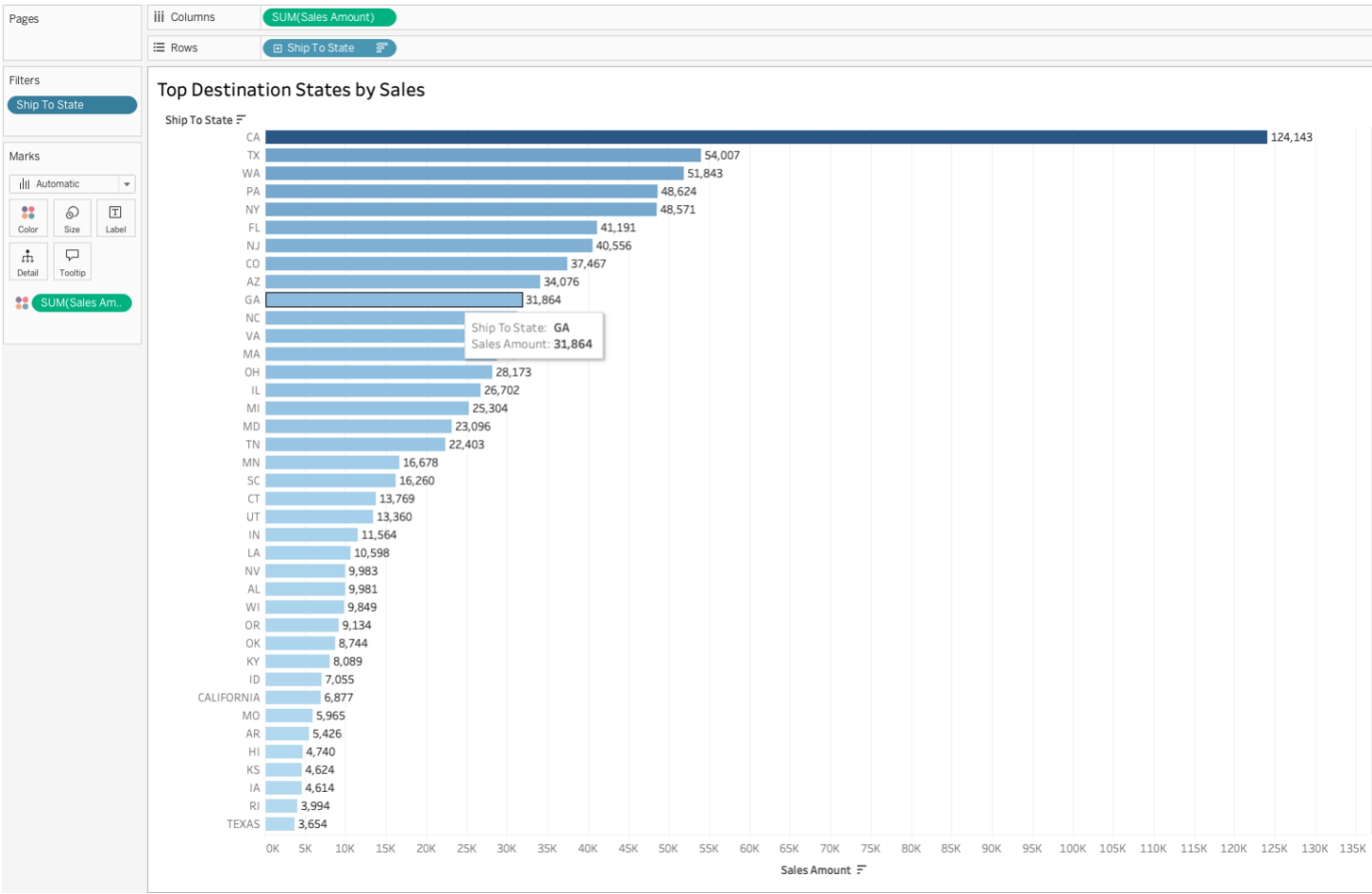
	year_num integer	month_num integer	ship_to_state text	order_count bigint	state_rank bigint
1	2020	3	NJ	2	1
2	2020	8	CA	2	1
3	2020	10	HI	2	1
4	2020	10	WA	2	1
5	2020	10	PA	1	2
6	2020	11	MARYLAND	6	1
7	2020	11	CA	3	2
8	2020	11	GA	3	2
9	2020	11	IL	3	2
10	2020	11	TN	2	3
11	2020	11	LA	1	4
12	2020	11	NV	1	4
13	2020	11	PA	1	4
14	2020	11	WI	1	4
15	2020	12	MD	6	1
16	2020	12	WA	6	1
17	2020	12	CA	4	2
18	2020	12	NY	4	2
19	2020	12	MN	3	3
20	2020	12	NC	3	3
21	2020	12	SC	3	3
22	2020	12	TX	3	3
23	2020	12	ALABAMA	2	4
24	2020	12	AR	2	4
25	2020	12	CO	2	4
26	2020	12	GEORGIA	2	4
27	2020	12	NJ	2	4
28	2020	12	WI	2	4
29	2020	12	WV	2	4
30	2020	12	AL	1	5
31	2020	12	AZ	1	5
32	2020	12	COLORADO	1	5
33	2020	12	CT	1	5
34	2020	12	GA	1	5

Key Insights

Inventory allocation, advertising investment, and regional growth strategies should prioritize the highest revenue states while identifying mid-tier states for expansion opportunities.

Final Conclusion

Inventory and marketing resources should prioritize the highest revenue states.

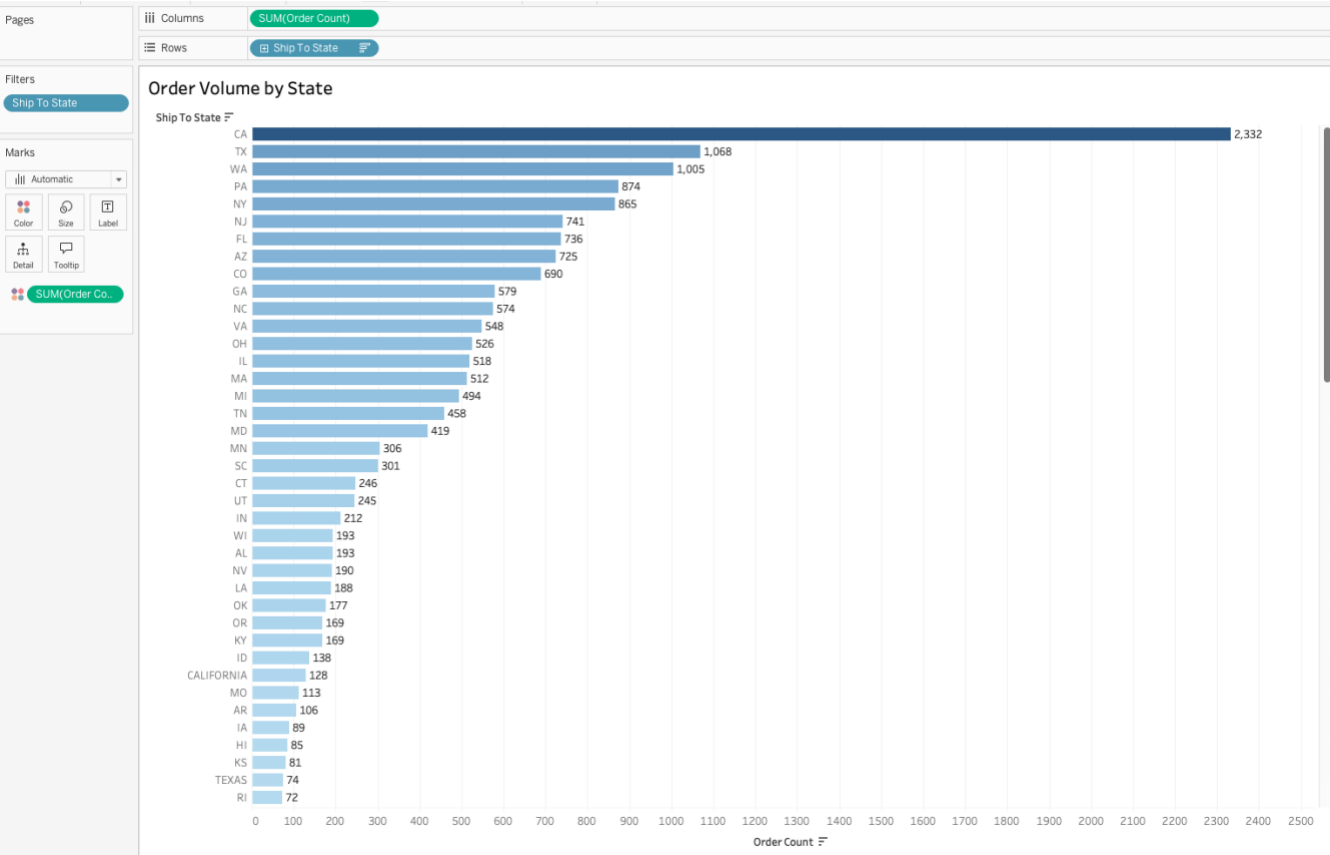


5. Business Question 2

Which states have the highest order volume?

Query Query History

```
32
33 select
34     dd.year_num,
35     dd.month_num,
36     dl.ship_to_state,
37     sum(f.sales_amount) as total_sales,
38     avg(f.sales_amount) as avg_order_value,
39     count(*) as order_count
40 from fact_order_snapshot f
41 join dim_date dd
42     on f.date_key = dd.date_key
43 join dim_destination_location dl
44     on f.destination_location_key = dl.destination_location_key
45 group by
46     dd.year_num,
47     dd.month_num,
48     dl.ship_to_state
49 order by
50     dd.year_num,
51     dd.month_num,
52     total_sales desc;
53
54
```



Data Output Messages Notifications							Showing rows:	
	year_num integer	month_num integer	ship_to_state text	total_sales numeric	avg_order_value numeric	order_count bigint		
1	2020	3	NJ	115.86	57.9300000000000000	2		
2	2020	8	CA	115.00	57.5000000000000000	2		
3	2020	10	HI	139.90	69.9500000000000000	2		
4	2020	10	WA	117.86	58.9300000000000000	2		
5	2020	10	PA	58.93	58.9300000000000000	1		
6	2020	11	MARYLAND	419.58	69.9300000000000000	6		
7	2020	11	CA	209.82	69.9400000000000000	3		
8	2020	11	IL	209.82	69.9400000000000000	3		
9	2020	11	TN	139.86	69.9300000000000000	2		
10	2020	11	GA	131.20	43.7333333333333333	3		
11	2020	11	NV	82.95	82.9500000000000000	1		
12	2020	11	WI	69.96	69.9600000000000000	1		
13	2020	11	LA	69.93	69.9300000000000000	1		
14	2020	11	PA	69.93	69.9300000000000000	1		
15	2020	12	MD	347.58	57.9300000000000000	6		
16	2020	12	NY	231.77	57.9425000000000000	4		
17	2020	12	CA	220.74	55.1850000000000000	4		
18	2020	12	MN	197.79	65.9300000000000000	3		
19	2020	12	WA	179.46	29.9100000000000000	6		
20	2020	12	GEORGIA	165.90	82.9500000000000000	2		
21	2020	12	NC	157.88	52.6266666666666667	3		
22	2020	12	SC	141.85	47.2833333333333333	3		
23	2020	12	WI	140.88	70.4400000000000000	2		
24	2020	12	AR	138.88	69.4400000000000000	2		
25	2020	12	TX	132.73	44.2433333333333333	3		
26	2020	12	WV	131.86	65.9300000000000000	2		
27	2020	12	ALABAMA	123.86	61.9300000000000000	2		
28	2020	12	CO	123.86	61.9300000000000000	2		
29	2020	12	NJ	117.87	58.9350000000000000	2		
30	2020	12	CT	82.95	82.9500000000000000	1		
31	2020	12	NV	69.93	69.9300000000000000	1		
32	2020	12	IL	69.93	69.9300000000000000	1		
33	2020	12	OH	69.93	69.9300000000000000	1		
34	2020	12	GA	65.93	65.9300000000000000	1		

Key Insights

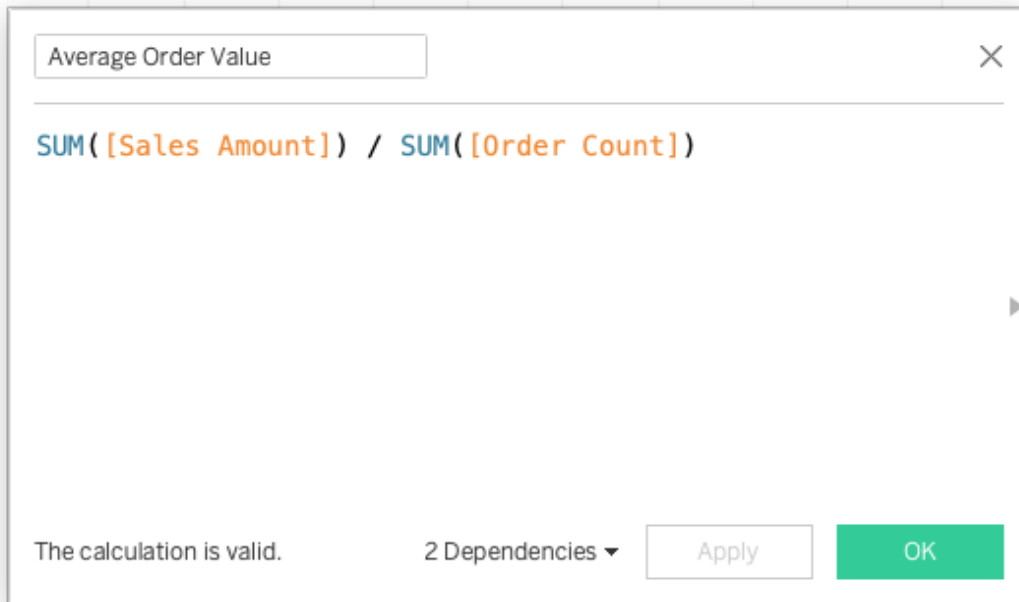
Order count should be analyzed together with sales revenue and average order value to support better market prioritization decisions.

Final Conclusion

Order count should be reviewed together with revenue and customer value.

6. Business Question 3

Which states have the highest average order value?



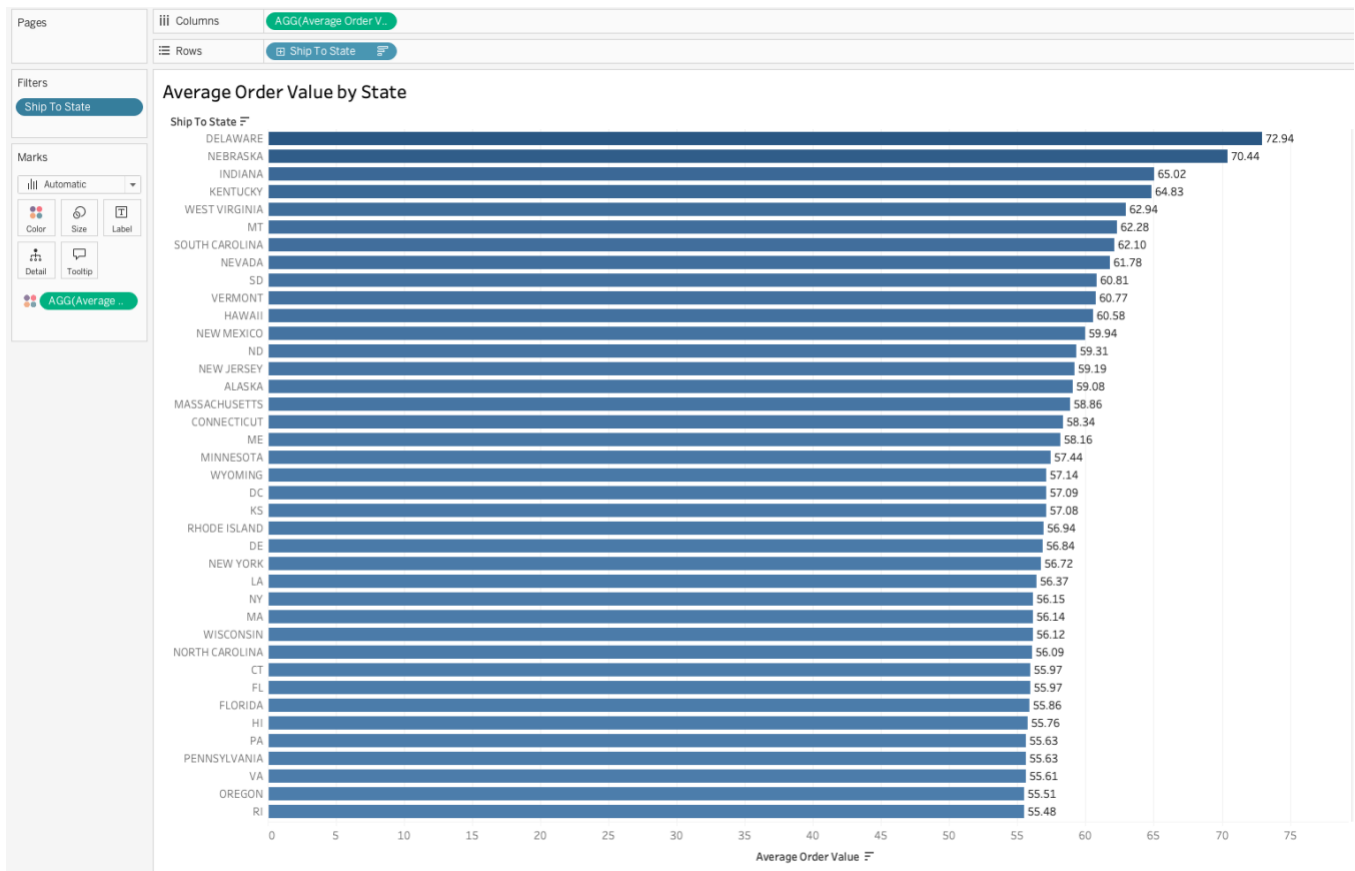
Average Order Value

`SUM([Sales Amount]) / SUM([Order Count])`

The calculation is valid. 2 Dependencies ▼

Apply OK

Average Order Value = $\text{SUM}(\text{Sales Amount}) / \text{SUM}(\text{Order Count})$



Key Insights

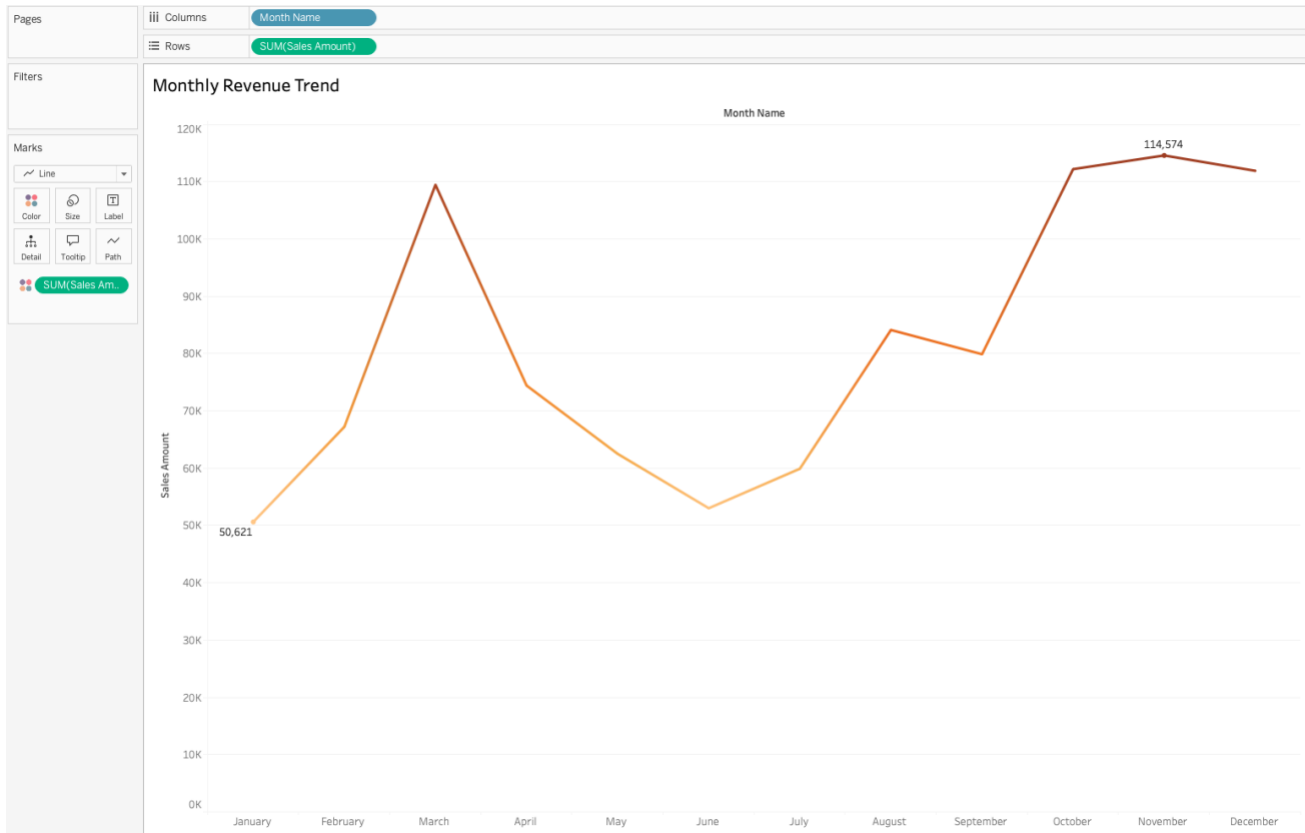
Premium pricing campaigns and targeted product promotions should focus on states with stronger average order value performance.

Final Conclusion

Premium marketing strategies should target high-value states.

7. Business Question 4

How does revenue change over time?



Key Insights

Inventory planning, staffing, and promotional campaigns should be expanded before Q4 to maximize seasonal demand opportunities.

Final Conclusion

Promotions and inventory should be expanded before Q4 seasonal demand.

8. Business Question 5

How do traffic and weather affect delivery performance?

```
84
85 select
86     dc.weather,
87     dc.traffic,
88     round(avg(f.delivery_time)::numeric, 2) as avg_delivery_time,
89     count(*) as delivery_count
90 from fact_delivery_cumulative f
91 join dim_condition dc
92     on f.condition_key = dc.condition_key
93 group by
94     dc.weather,
95     dc.traffic
96 order by
97     avg_delivery_time desc,
98     delivery_count desc;
99
```

Data Output Messages Notifications				
	weather text	traffic text	avg_delivery_time numeric	delivery_count bigint
1	Cloudy	Jam	174.58	2267
2	Fog	Jam	174.16	2365
3	Windy	Jam	142.93	2286
4	Sandstor...	Jam	142.30	2316
5	Stormy	Jam	142.29	2259
6	Cloudy	High	138.74	717
7	Cloudy	Medium	136.68	1779
8	Fog	High	134.94	757
9	Sandstor...	Medium	132.86	1731
10	Fog	Medium	132.19	1798
11	Stormy	High	131.92	713
12	Sandstor...	High	131.86	678
13	Stormy	Medium	131.80	1783
14	Windy	Medium	130.65	1808
15	Windy	High	128.73	715
16	Nan	Nan	120.67	91
17	Sunny	High	110.18	716
18	Sunny	Jam	108.68	2232
19	Cloudy	Low	106.70	2525
20	Fog	Low	104.90	2520
21	Sunny	Low	102.57	2401
22	Windy	Low	98.67	2414
23	Stormy	Low	98.54	2619
24	Sandstor...	Low	96.79	2520
25	Sunny	Medium	96.02	1729

Key Insights

Delivery staffing levels, route planning, and operational scheduling should adapt dynamically based on traffic and weather forecasts.

Final Conclusion

Delivery staffing and routing should adjust based on traffic and weather forecasts.

9. Slowly Changing Dimensions

SCD Type 2 – dim_destination_location

Tracks historical destination changes by expiring old records and inserting new active rows while preserving prior history.

```
2
3 create table destination_location_delta (
4     ship_to_country text,
5     ship_to_state text,
6     ship_to_city text,
7     ship_to_postal_code text
8 );
9
10 insert into destination_location_delta
11 values
12 ('US', 'CA', 'Los Angeles', '90001'),
13 ('US', 'TX', 'Houston', '77001'),
14 ('US', 'NY', 'Buffalo', '14201');
15
16 select *
17 from dim_destination_location
18 where (ship_to_state = 'CA' and ship_to_city = 'Los Angeles')
19       or (ship_to_state = 'TX' and ship_to_city = 'Houston')
20       or (ship_to_state = 'NY' and ship_to_city = 'Buffalo')
21 order by ship_to_state, ship_to_city, start_date;
22
```

	destination_location_key [PK] bigint	ship_to_country text	ship_to_state text	ship_to_city text	ship_to_postal_code text	start_date date	end_date date	current_flag character (1)
1	10538	US	CA	Los Angeles	90001	2026-04-20	[null]	Y
2	10080	US	CA	Los Angeles	90027-2108	2026-04-20	2026-04-20	N
3	10028	US	CA	Los Angeles	90066-2844	2026-04-20	2026-04-20	N
4	9770	US	CA	Los Angeles	90020-4616	2026-04-20	2026-04-20	N
5	9259	US	CA	Los Angeles	90042-2230	2026-04-20	2026-04-20	N
6	8989	US	CA	Los Angeles	90043-5342	2026-04-20	2026-04-20	N
7	8583	US	CA	Los Angeles	90046-2970	2026-04-20	2026-04-20	N
8	8341	US	CA	Los Angeles	90025-6538	2026-04-20	2026-04-20	N
9	7975	US	CA	Los Angeles	90016	2026-04-20	2026-04-20	N
10	7656	US	CA	Los Angeles	90026-1913	2026-04-20	2026-04-20	N
11	7007	US	CA	Los Angeles	90063-2619	2026-04-20	2026-04-20	N
12	6049	US	CA	Los Angeles	90046-7739	2026-04-20	2026-04-20	N
13	5983	US	CA	Los Angeles	90039-2537	2026-04-20	2026-04-20	N
14	5930	US	CA	Los Angeles	90043-1948	2026-04-20	2026-04-20	N
15	5843	US	CA	Los Angeles	90045-1320	2026-04-20	2026-04-20	N
16	5314	US	CA	Los Angeles	90066-1817	2026-04-20	2026-04-20	N
17	5300	US	CA	Los Angeles	90026-4383	2026-04-20	2026-04-20	N
18	4846	US	CA	Los Angeles	90066-5616	2026-04-20	2026-04-20	N
19	4793	US	CA	Los Angeles	90059-2822	2026-04-20	2026-04-20	N
20	4774	US	CA	Los Angeles	90005-4080	2026-04-20	2026-04-20	N
21	4686	US	CA	Los Angeles	90045-2618	2026-04-20	2026-04-20	N
22	4389	US	CA	Los Angeles	90031-2589	2026-04-20	2026-04-20	N
23	4368	US	CA	Los Angeles	90068-1724	2026-04-20	2026-04-20	N
24	3521	US	CA	Los Angeles	90045-2530	2026-04-20	2026-04-20	N
25	2887	US	CA	Los Angeles	90041-1532	2026-04-20	2026-04-20	N
26	1913	US	CA	Los Angeles	90035-4209	2026-04-20	2026-04-20	N
27	1837	US	CA	Los Angeles	90073-1003	2026-04-20	2026-04-20	N
28	1550	US	CA	Los Angeles	90046-1418	2026-04-20	2026-04-20	N
29	1483	US	CA	Los Angeles	90039-2648	2026-04-20	2026-04-20	N
30	601	US	CA	Los Angeles	90045-3130	2026-04-20	2026-04-20	N
31	196	US	CA	Los Angeles	90046	2026-04-20	2026-04-20	N
32	10206	US	CA	Los Angeles	90019-2032	2026-04-20	2026-04-20	N
33	10540	US	NY	Buffalo	14201	2026-04-20	[null]	Y
34	9244	US	NY	Buffalo	14213-2153	2026-04-20	2026-04-20	N

```

29 from destination_location_delta x
30 where d.ship_to_country = x.ship_to_country
31      and d.ship_to_state = x.ship_to_state
32      and d.ship_to_city = x.ship_to_city
33      and d.current_flag = 'Y'
34      and d.ship_to_postal_code <> x.ship_to_postal_code;
35
36
37
38 -- insert new current record
39
40 insert into dim_destination_location
41 (
42     ship_to_country,
43     ship_to_state,
44     ship_to_city,
45     ship_to_postal_code,
46     start_date,
47     end_date,
48     current_flag
49 )
50 select
51     x.ship_to_country,
52     x.ship_to_state,
53     x.ship_to_city,
54     x.ship_to_postal_code,
55     current_date,
56     null,
57     'Y'
58 from destination_location_delta x
59 left join dim_destination_location d
60     on x.ship_to_country = d.ship_to_country
61     and x.ship_to_state = d.ship_to_state
62     and x.ship_to_city = d.ship_to_city
63     and x.ship_to_postal_code = d.ship_to_postal_code
64     and d.current_flag = 'Y'
65 where d.destination_location_key is null;
66
67
68
69 -- check scd2 result
70
71 select *
72 from dim_destination_location
73 order by ship_to_state, ship_to_city, start_date;
74

```

	destination_location_key [PK] bigint	ship_to_country text	ship_to_state text	ship_to_city text	ship_to_postal_code text	start_date date	end_date date	current_flag character (1)
1	6713	US	AK	Anchorage	99503-5774	2026-04-20	[null]	Y
2	7562	US	AK	Anchorage	99504-2317	2026-04-20	[null]	Y
3	4302	US	AK	Anchorage	99508-5021	2026-04-20	[null]	Y
4	7152	US	AK	Anchorage	99508	2026-04-20	[null]	Y
5	2324	US	AK	Anchorage	99504-3405	2026-04-20	[null]	Y
6	1422	US	AK	Delta Junction	99737-0881	2026-04-20	[null]	Y
7	3902	US	AK	Eagle River	99577-8504	2026-04-20	[null]	Y
8	9040	US	AK	Eagle River	99577-7798	2026-04-20	[null]	Y
9	9853	US	AK	Juneau	99801-2015	2026-04-20	[null]	Y
10	3194	US	AK	Kenai	99611-8417	2026-04-20	[null]	Y
11	5815	US	AK	Palmer	99645-7700	2026-04-20	[null]	Y
12	10145	US	AK	Palmer	99645-8940	2026-04-20	[null]	Y
13	8579	US	AK	Wasilla	99687	2026-04-20	[null]	Y
14	4772	US	AK	Wasilla	99623-4916	2026-04-20	[null]	Y
15	5481	US	AK	Wasilla	99623-0767	2026-04-20	[null]	Y
16	4564	US	AK	Wasilla	99654	2026-04-20	[null]	Y
17	7634	US	AL	Albertville	35951-7348	2026-04-20	[null]	Y
18	7773	US	AL	Anniston	36207-5008	2026-04-20	[null]	Y
19	1393	US	AL	Anniston	36207-1184	2026-04-20	[null]	Y
20	5388	US	AL	Athens	35613-6630	2026-04-20	[null]	Y
21	6022	US	AL	Auburn	36832-8668	2026-04-20	[null]	Y
22	8386	US	AL	Auburn	36830-6831	2026-04-20	[null]	Y
23	9484	US	AL	Auburn	36830-0335	2026-04-20	[null]	Y
24	8984	US	AL	Auburn	36830-0244	2026-04-20	[null]	Y
25	1947	US	AL	Bay Minette	36507	2026-04-20	[null]	Y
26	5229	US	AL	Birmingham	35242-3450	2026-04-20	[null]	Y
27	1193	US	AL	Birmingham	35244-2285	2026-04-20	[null]	Y
28	8327	US	AL	Birmingham	35244-1600	2026-04-20	[null]	Y
29	311	US	AL	Birmingham	35206-3954	2026-04-20	[null]	Y
30	6809	US	AL	Birmingham	35242-4402	2026-04-20	[null]	Y
31	3340	US	AL	Birmingham	35233-2705	2026-04-20	[null]	Y
32	3483	US	AL	Clanton	35046-4612	2026-04-20	[null]	Y
33	5280	US	AL	Columbiana	35051-3940	2026-04-20	[null]	Y
34	1635	US	AL	Cullman	35055-6237	2026-04-20	[null]	Y

SCD Type 3 – dim_fulfillment

Tracks previous and current fulfillment values within the same row to preserve recent attribute changes.

```
78
79 create table fulfillment_delta (
80     new_fulfillment_type text,
81     old_fulfillment_type text
82 );
83
84 insert into fulfillment_delta
85 values
86 ('AMAZON_FRESH', 'AMAZON'),
87 ('MERCHANT_FULFILLED', 'SELLER');
88
89 select *
90 from dim_fulfillment
91 where current_fulfillment_type in ('AFN', 'MFN')
92 order by fulfillment_key;
93
94
```

Data Output Messages Notifications

        SQL

	fulfillment_key [PK] bigint	current_fulfillment_type text	previous_fulfillment_type text
1	1	AFN	[null]
2	2	MFN	[null]

```

95
96 update dim_fulfillment f
97 set
98     previous_fulfillment_type = f.current_fulfillment_type,
99     current_fulfillment_type = d.new_fulfillment_type
100 from fulfillment_delta d
101 where f.current_fulfillment_type = d.old_fulfillment_type;
102
103 |
104 -- check scd3 result
105
106 select *
107 from dim_fulfillment
108 order by fulfillment_key;

```

Data Output Messages Notifications







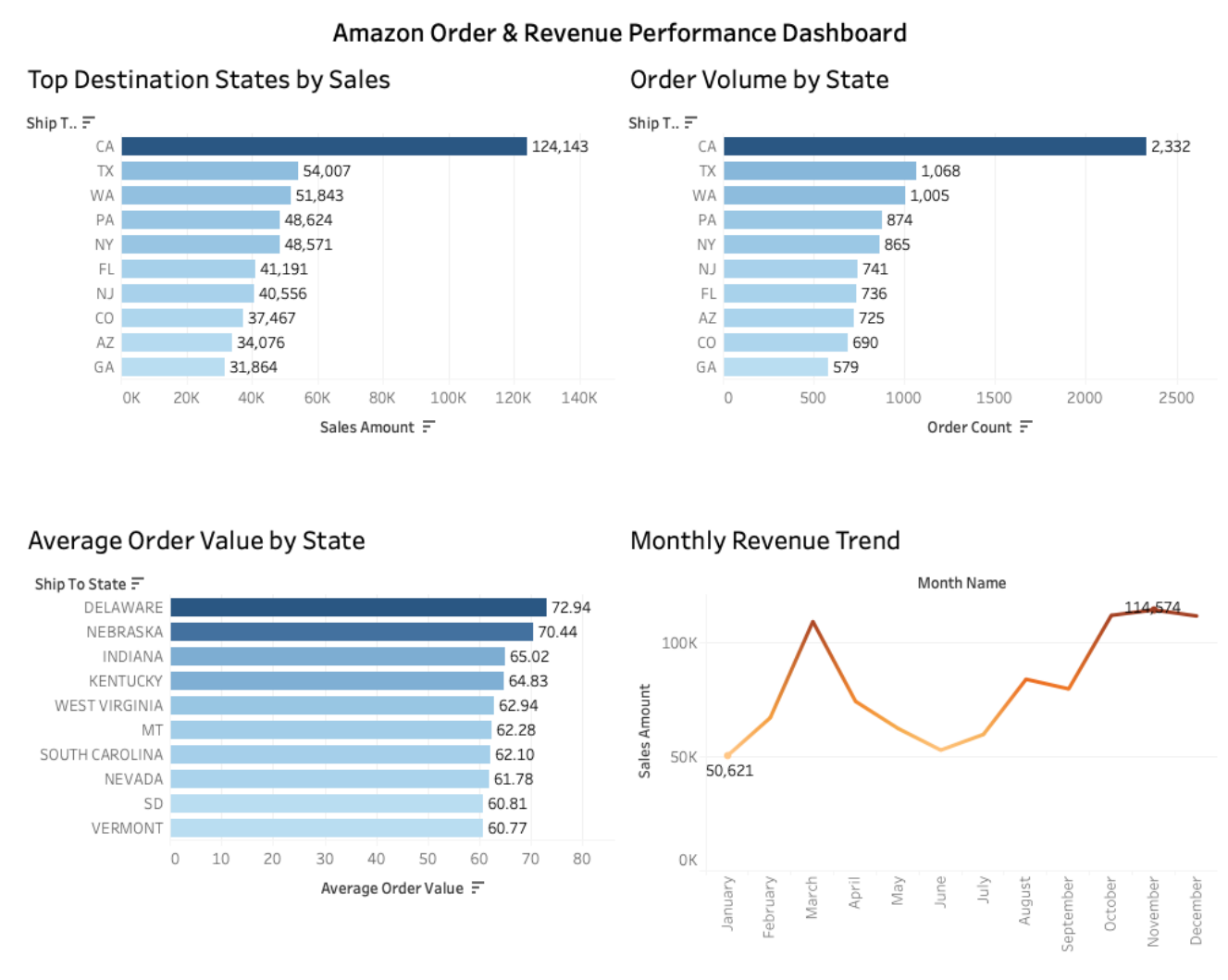



SQL

	fulfillment_key [PK] bigint	current_fulfillment_type text	previous_fulfillment_type text
1	1	AFN_UPDATED	AFN
2	2	MFN_UPDATED	MFN

10.Final Dashboard

The final Tableau dashboard was enhanced to improve usability, interactivity, and executive reporting quality.



Dashboard Features Completed

- At least three visualizations
- Drill-down hierarchy
- Filtering
- Calculated field
- Multiple chart types

11.Source Code Included

 1_staging_and_checks.sql	Today
 2_cleaned_tables.sql	Today
 3_dimensions_facts_validation.sql	Today
 4_business_queries.sql	Today
 5_scd_examples.sql	Today

All SQL source code files were included in the final submission package.

Files included:

1_staging_and_checks.sql

2_cleaned_tables.sql

3_dimensions_fact.sql

4_business_queries.sql

5_scd_examples.sql

12.Final Reflection

This project significantly improved my understanding of dimensional modeling, ETL development, SQL analytics, and business intelligence reporting.

I learned how raw operational data can be transformed into structured fact and dimension tables that support business decision-making.

I also gained practical experience implementing Slowly Changing Dimensions, designing Tableau dashboards, and translating business questions into measurable KPIs.

Overall, this project helped connect technical warehouse design with real-world analytics strategy and decision support.